MH2500 PROBABILITY AND INTRODUCTION TO STATISTICS

NANYANG TECHNOLOGICAL UNIVERSITY

Tutorial 2 HINTS

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We will discuss several questions from Chapters 2 and 3 in Ross' book.

Tutorial Questions

(1) In an experiment, die is rolled continually until a 6 appears, at which point the experiment stops. What is the sample space of this experiment? Let E_n denote the event that n rolls are necessary to complete the experiment. What points of the sample space are contained in E_n? What is $\left(\bigcup_{i=1}^{\infty} E_n\right)'$?

Answer: The sample space consists of the following sequences of die rolls. If the sequence has finite length, then its last entry must equal six and no other entry can be six. If the sequence is infinite, then no entry can equal six.

The set E_n consists of all sequences of length n in the sample space.

The set $\left(\bigcup_{i=1}^{\infty} E_n\right)'$ consists of all sequences of infinite length in the sample space.

(Note: $\left(\bigcup_{i=1}^{\infty} E_n\right)'$ is uncountable.)

(2) If 8 rooks (castles) are randomly placed on a chessboard, compute the probability that none of the rooks can capture any of the others. That is, compute the probability that no row or file contains more than one rook.

Answer:

$$\frac{64 \cdot 49 \cdot 36 \cdot 25 \cdot 16 \cdot 9 \cdot 4 \cdot 1}{64 \cdot 63 \cdots 58 \cdot 57} = \frac{560}{61474519} \approx 9.109 \times 10^{-6}.$$

(3) There are n socks, 3 of which are red, in a drawer. What is the value of n if, when 2 of the socks are chosen randomly, the probability that they are both red is 1/2?

Answer: By

$$\frac{\binom{3}{2}}{\binom{n}{2}} = \frac{1}{2},$$

we have $n = 4$.

(4) If $P(E) = 0.9$ and $P(F) = 0.8$, show that $P(E \cap F) \geq 0.7$. In general, prove Bonferroni's inequality, namely,

$$P(E \cap F) \geq P(E) + P(F) - 1.$$

Proof:

$$\begin{aligned} P(E \cap F) = 1 - P((E \cap F)') &= 1 - P(E' \cup F') \\ &\geq 1 - P(E') - P(F') \\ &= 1 - (1 - P(E)) - (1 - P(F)) \\ &= P(E) + P(F) - 1. \end{aligned}$$

Now $P(E \cap F) \geq 0.7$ if $P(E) = 0.9$ and $P(F) = 0.8$.

(5) Show that if $P(A_i) = 1$ for all $i \geq 1$ then $P\left(\bigcap_{i=1}^{\infty} A_i\right) = 1$.

Proof: We look at the complement of the events.

$$P\left(\left(\bigcap_{i=1}^{\infty} A_i\right)'\right) = P\left(\bigcup_{i=1}^{\infty} A_i'\right).$$

Let $B_n = \bigcup_{i=1}^{n} A_i'$. Then $B_n \subseteq B_{n+1}$,

$$P(B_n) = P\left(\bigcup_{i=1}^{n} A_i'\right) \leq \sum_{i=1}^{n} P(A_i') = 0,$$

and $\bigcup_{i=1}^{n} B_i = \bigcup_{i=1}^{n} A_i'$. This gives $\bigcup_{i=1}^{\infty} A_i' = \lim_{n\to\infty} \bigcup_{i=1}^{n} A_i' = \lim_{n\to\infty} \bigcup_{i=1}^{n} B_i = \bigcup_{i=1}^{\infty} B_i$. As $\{B_n\}$ is an increasing sequence sets, we have

$$P\left(\bigcup_{i=1}^{\infty} B_i\right) = \lim_{i\to\infty} P(B_i) = 0$$

and hence

$$P\left(\bigcup_{i=1}^{\infty} A_i'\right) = P\left(\bigcup_{i=1}^{\infty} B_i\right) = 0.$$

That is,

$$P\left(\left(\bigcap_{i=1}^{\infty} A_i\right)'\right) = 0, \quad \text{and hence} \quad P\left(\bigcap_{i=1}^{\infty} A_i\right) = 1.$$

(6) Celine is undecided as to whether to take a French course or a chemistry course. She estimates that her probability of receiving an A grade would be $1/2$ in a French course and $2/3$ in a chemistry course. If Celine decides to base her decision on the flip of a fair coin, what is the probability that she gets an A in chemistry?

Hint: Let C denote the event that Celine takes chemistry and A denote the event that she receives an A in whatever course she takes, then

$$P(A|C) = 2/3, \qquad P(C) = 1/2,$$

and the desired probability $P(CA)$ is calculated as follows:

$$P(CA) = P(C)P(A|C) = \left(\frac{1}{2}\right)\left(\frac{2}{3}\right) = \frac{1}{3}.$$

(7) Three cards are randomly selected, without replacement, from an ordinary deck of 52 playing cards. Compute the conditional probability that the first card selected is a spade given that the second and third cards are spades.

Hint:

$$\begin{aligned}
&P(\text{1st is spade} \mid \text{2nd and 3rd are spades})\\
&= \frac{P(\text{all three cards are spades})}{P(\text{2nd and 3rd are spades})}\\
&= \frac{P(\text{all three cards are spades})}{P(\text{all three cards are spades}) + P(\text{1st not spade, 2nd and 3rd are spades})}\\
&= \frac{\frac{13}{52}\cdot\frac{12}{51}\cdot\frac{11}{50}}{\frac{13}{52}\cdot\frac{12}{51}\cdot\frac{11}{50}+\frac{39}{52}\cdot\frac{13}{51}\cdot\frac{12}{50}}\\
&= \frac{11}{50}.
\end{aligned}$$

(8) A laboratory blood test is 95 percent effective in detecting a certain disease when it is, in fact, present. However, the test also yields a "false positive" result for 1 percent of the healthy persons tested. (That is, if a healthy person is tested, then, with probability .01, the test result will imply that he or she has the disease.) If .5 percent of the population actually has the disease, what is the probability that a person has the disease given that the test result is positive?

Hint: Let D be the event that the person tested has the disease and $+$ the event that the test result is positive. Then $P(D) = .005$, $P(D') = 0.995$, $P(+|D) = .95$, $P(+|D') = .01$, and, by Bayes' formula, the desired probability is

$$\begin{aligned}
P(D|+) &= \frac{P(+|D)P(D)}{P(+|D)P(D) + P(+|D')P(D')}\\
&= \frac{0.95\cdot 0.005}{0.95\cdot 0.005 + 0.01\cdot 0.995} = \frac{95}{294} \approx 0.323.
\end{aligned}$$